**11. On Example 4.** Show that in Example 4 of the text, $\mathbf{F} = \text{grad}\,(\arctan\,(y/x))$. Give examples of domains in which the integral is path independent.

**12. CAS EXPERIMENT. Extension of Project 10.** Integrate $x^2y\,dx + 2xy^2\,dy$ over various circles through the points $(0, 0)$ and $(1, 1)$. Find experimentally the smallest value of the integral and the approximate location of the center of the circle.

**13–19 PATH INDEPENDENCE?**

Check, and if independent, integrate from $(0, 0, 0)$ to $(a, b, c)$.

**13.** $2e^{x^2}(x \cos 2y\,dx - \sin 2y\,dy)$

**14.** $(\sinh xy)\,(z\,dx - x\,dz)$

**15.** $x^2y\,dx - 4xy^2\,dy + 8z^2x\,dz$

**16.** $e^y\,dx + (xe^y - e^z)\,dy - ye^z\,dz$

**17.** $4y\,dx + z\,dy + (y - 2z)\,dz$

**18.** $(\cos xy)(yz\,dx + xz\,dy) - 2 \sin xy\,dz$

**19.** $(\cos\,(x^2 + 2y^2 + z^2))\,(2x\,dx + 4y\,dy + 2z\,dz)$

**20. Path Dependence.** Construct three simple examples in each of which two equations $(6')$ are satisfied, but the third is not.

# 10.3 Calculus Review: Double Integrals. *Optional*

*This section is optional. Students familiar with double integrals from calculus should skip this review and go on to Sec. 10.4.* This section is included in the book to make it reasonably self-contained.

In a definite integral (1), Sec. 10.1, we integrate a function $f(x)$ over an interval (a segment) of the $x$-axis. In a double integral we integrate a function $f(x, y)$, called the *integrand,* over a closed bounded region[2] $R$ in the $xy$-plane, whose boundary curve has a unique tangent at almost every point, but may perhaps have finitely many cusps (such as the vertices of a triangle or rectangle).

The definition of the double integral is quite similar to that of the definite integral. We subdivide the region $R$ by drawing parallels to the $x$- and $y$-axes (Fig. 227). We number the rectangles that are entirely within $R$ from 1 to $n$. In each such rectangle we choose a point, say, $(x_k, y_k)$ in the $k$th rectangle, whose area we denote by $\Delta A_k$. Then we form the sum

$$J_n = \sum_{k=1}^{n} f(x_k, y_k)\,\Delta A_k.$$

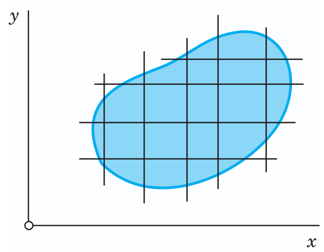


Fig. 227. Subdivision of a region $R$

[2]A **region** $R$ is a domain (Sec. 9.6) plus, perhaps, some or all of its boundary points. $R$ is **closed** if its **boundary** (all its boundary points) are regarded as belonging to $R$; and $R$ is **bounded** if it can be enclosed in a circle of sufficiently large radius. A **boundary point** $P$ of $R$ is a point (of $R$ or not) such that every disk with center $P$ contains points of $R$ and also points not of $R$.

This we do for larger and larger positive integers $n$ in a completely independent manner, but so that the length of the maximum diagonal of the rectangles approaches zero as $n$ approaches infinity. In this fashion we obtain a sequence of real numbers $J_{n_1}, J_{n_2}, \cdots$. Assuming that $f(x, y)$ is continuous in $R$ and $R$ is bounded by finitely many smooth curves (see Sec. 10.1), one can show (see Ref. [GenRef4] in App. 1) that this sequence converges and its limit is independent of the choice of subdivisions and corresponding points $(x_k, y_k)$. This limit is called the **double integral** *of* $f(x, y)$ *over the region* $R$, and is denoted by

$$\iint_R f(x, y)\, dx\, dy \qquad \text{or} \qquad \iint_R f(x, y)\, dA.$$

Double integrals have properties quite similar to those of definite integrals. Indeed, for any functions $f$ and $g$ of $(x, y)$, defined and continuous in a region $R$,

$$\iint_R kf\, dx\, dy = k \iint_R f\, dx\, dy \qquad (k \text{ constant})$$

(1)
$$\iint_R (f + g)\, dx\, dy = \iint_R f\, dx\, dy + \iint_R g\, dx\, dy$$

$$\iint_R f\, dx\, dy = \iint_{R_1} f\, dx\, dy + \iint_{R_2} f\, dx\, dy \qquad \text{(Fig. 228).}$$

Furthermore, if $R$ is simply connected (see Sec. 10.2), then there exists at least one point $(x_0, y_0)$ in $R$ such that we have

(2)
$$\iint_R f(x, y)\, dx\, dy = f(x_0, y_0)A,$$

where $A$ is the area of $R$. This is called the **mean value theorem** *for double integrals.*

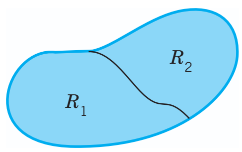


**Fig. 228.** Formula (1)

## Evaluation of Double Integrals by Two Successive Integrations

*Double integrals over a region* $R$ *may be evaluated by two successive integrations.* We may integrate first over $y$ and then over $x$. Then the formula is

**(3)**
$$\boxed{\iint_R f(x, y)\, dx\, dy = \int_a^b \left[ \int_{g(x)}^{h(x)} f(x, y)\, dy \right] dx} \qquad \text{(Fig. 229).}$$

Here $y = g(x)$ and $y = h(x)$ represent the boundary curve of $R$ (see Fig. 229) and, keeping $x$ constant, we integrate $f(x, y)$ over $y$ from $g(x)$ to $h(x)$. The result is a function of $x$, and we integrate it from $x = a$ to $x = b$ (Fig. 229).

Similarly, for integrating first over $x$ and then over $y$ the formula is

$$\iint_R f(x, y)\, dx\, dy = \int_c^d \left[ \int_{p(y)}^{q(y)} f(x, y)\, dx \right] dy \tag{4}$$

(Fig. 230).

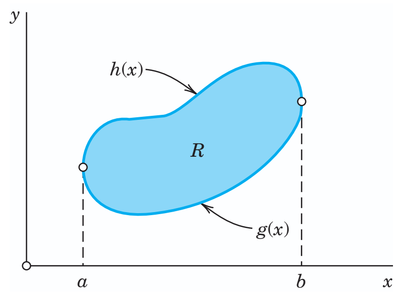


**Fig. 229.** Evaluation of a double integral

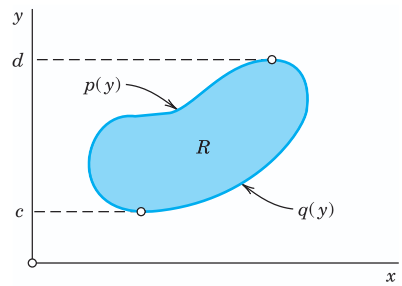


**Fig. 230.** Evaluation of a double integral

The boundary curve of $R$ is now represented by $x = p(y)$ and $x = q(y)$. Treating $y$ as a constant, we first integrate $f(x, y)$ over $x$ from $p(y)$ to $q(y)$ (see Fig. 230) and then the resulting function of $y$ from $y = c$ to $y = d$.

In (3) we assumed that $R$ can be given by inequalities $a \leqq x \leqq b$ and $g(x) \leqq y \leqq h(x)$. Similarly in (4) by $c \leqq y \leqq d$ and $p(y) \leqq x \leqq q(y)$. If a region $R$ has no such representation, then, in any practical case, it will at least be possible to subdivide $R$ into finitely many portions each of which can be given by those inequalities. Then we integrate $f(x, y)$ over each portion and take the sum of the results. This will give the value of the integral of $f(x, y)$ over the entire region $R$.

## Applications of Double Integrals

Double integrals have various physical and geometric applications. For instance, the **area** $A$ of a region $R$ in the $xy$-plane is given by the double integral

$$A = \iint_R dx\, dy.$$

The **volume** $V$ beneath the surface $z = f(x, y)$ $(>0)$ and above a region $R$ in the $xy$-plane is (Fig. 231)

$$V = \iint_R f(x, y)\, dx\, dy$$

because the term $f(x_k, y_k)\Delta A_k$ in $J_n$ at the beginning of this section represents the volume of a rectangular box with base of area $\Delta A_k$ and altitude $f(x_k, y_k)$.

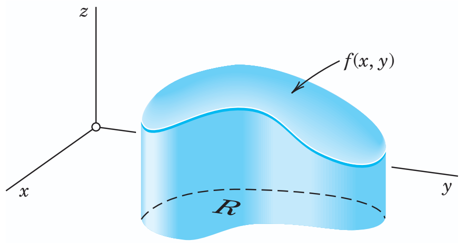


**Fig. 231.** Double integral as volume

As another application, let $f(x, y)$ be the density (= mass per unit area) of a distribution of mass in the $xy$-plane. Then the **total mass** $M$ in $R$ is

$$M = \iint_R f(x, y)\, dx\, dy;$$

the **center of gravity** of the mass in $R$ has the coordinates $\bar{x}, \bar{y}$, where

$$\bar{x} = \frac{1}{M}\iint_R x f(x, y)\, dx\, dy \qquad \text{and} \qquad \bar{y} = \frac{1}{M}\iint_R y f(x, y)\, dx\, dy;$$

the **moments of inertia** $I_x$ and $I_y$ of the mass in $R$ about the $x$- and $y$-axes, respectively, are

$$I_x = \iint_R y^2 f(x, y)\, dx\, dy, \qquad I_y = \iint_R x^2 f(x, y)\, dx\, dy;$$

and the **polar moment of inertia** $I_0$ about the origin of the mass in $R$ is

$$I_0 = I_x + I_y = \iint_R (x^2 + y^2) f(x, y)\, dx\, dy.$$

An example is given below.

## Change of Variables in Double Integrals. Jacobian

Practical problems often require a change of the variables of integration in double integrals. Recall from calculus that for a definite integral the formula for the change from $x$ to $u$ is

(5) $$\int_a^b f(x)\, dx = \int_\alpha^\beta f(x(u))\, \frac{dx}{du}\, du.$$

Here we assume that $x = x(u)$ is continuous and has a continuous derivative in some interval $\alpha \leqq u \leqq \beta$ such that $x(\alpha) = a$, $x(\beta) = b$ [or $x(\alpha) = b$, $x(\beta) = a$] and $x(u)$ varies between $a$ and $b$ when $u$ varies between $\alpha$ and $\beta$.

The formula for a change of variables in double integrals from $x, y$ to $u, v$ is

**(6)** $$\iint_R f(x, y)\, dx\, dy = \iint_{R^*} f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du\, dv;$$

that is, the integrand is expressed in terms of $u$ and $v$, and $dx\,dy$ is replaced by $du\,dv$ times the absolute value of the **Jacobian**[3]

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}. \tag{7}$$

Here we assume the following. The functions

$$x = x(u, v), \qquad y = y(u, v)$$

effecting the change are continuous and have continuous partial derivatives in some region $R^*$ in the $uv$-plane such that for every $(u, v)$ in $R^*$ the corresponding point $(x, y)$ lies in $R$ and, conversely, to every $(x, y)$ in $R$ there corresponds one and only one $(u, v)$ in $R^*$; furthermore, the Jacobian $J$ is either positive throughout $R^*$ or negative throughout $R^*$. For a proof, see Ref. [GenRef4] in App. 1.

**EXAMPLE 1 Change of Variables in a Double Integral**

Evaluate the following double integral over the square $R$ in Fig. 232.

$$\iint_R (x^2 + y^2)\,dx\,dy$$

***Solution.*** The shape of $R$ suggests the transformation $x + y = u, x - y = v$. Then $x = \frac{1}{2}(u + v)$, $y = \frac{1}{2}(u - v)$. The Jacobian is

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{vmatrix} = -\frac{1}{2}.$$

$R$ corresponds to the square $0 \leqq u \leqq 2, 0 \leqq v \leqq 2$. Therefore,

$$\iint_R (x^2 + y^2)\,dx\,dy = \int_0^2\int_0^2 \frac{1}{2}(u^2 + v^2)\frac{1}{2}\,du\,dv = \frac{8}{3}.$$ ∎

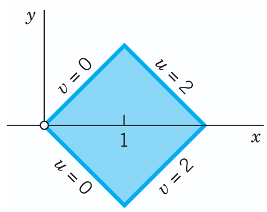


Fig. 232. Region $R$ in Example 1

[3]Named after the German mathematician CARL GUSTAV JACOB JACOBI (1804–1851), known for his contributions to elliptic functions, partial differential equations, and mechanics.

Of particular practical interest are **polar coordinates** $r$ and $\theta$, which can be introduced by setting $x = r\cos\theta$, $\quad y = r\sin\theta$. Then

$$J = \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r$$

and

$$\iint_R f(x, y)\,dx\,dy = \iint_{R^*} f(r\cos\theta, r\sin\theta)\,r\,dr\,d\theta \tag{8}$$

where $R^*$ is the region in the $r\theta$-plane corresponding to $R$ in the $xy$-plane.

**EXAMPLE 2** **Double Integrals in Polar Coordinates. Center of Gravity. Moments of Inertia**

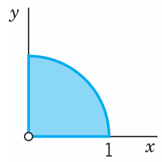

**Fig. 233.** Example 2

Let $f(x, y) = 1$ be the mass density in the region in Fig. 233. Find the total mass, the center of gravity, and the moments of inertia $I_x$, $I_y$, $I_0$.

***Solution.*** We use the polar coordinates just defined and formula (8). This gives the total mass

$$M = \iint_R dx\,dy = \int_0^{\pi/2}\int_0^1 r\,dr\,d\theta = \int_0^{\pi/2} \frac{1}{2}\,d\theta = \frac{\pi}{4}.$$

The center of gravity has the coordinates

$$\bar{x} = \frac{4}{\pi}\int_0^{\pi/2}\int_0^1 r\cos\theta\, r\,dr\,d\theta = \frac{4}{\pi}\int_0^{\pi/2} \frac{1}{3}\cos\theta\,d\theta = \frac{4}{3\pi} = 0.4244$$

$$\bar{y} = \frac{4}{3\pi} \qquad \text{for reasons of symmetry.}$$

The moments of inertia are

$$I_x = \iint_R y^2\,dx\,dy = \int_0^{\pi/2}\int_0^1 r^2\sin^2\theta\, r\,dr\,d\theta = \int_0^{\pi/2} \frac{1}{4}\sin^2\theta\,d\theta$$

$$= \int_0^{\pi/2} \frac{1}{8}(1 - \cos 2\theta)\,d\theta = \frac{1}{8}\left(\frac{\pi}{2} - 0\right) = \frac{\pi}{16} = 0.1963$$

$$I_y = \frac{\pi}{16} \quad \text{for reasons of symmetry,} \qquad I_0 = I_x + I_y = \frac{\pi}{8} = 0.3927.$$

Why are $\bar{x}$ and $\bar{y}$ less than $\frac{1}{2}$? ■

This is the end of our review on double integrals. These integrals will be needed in this chapter, beginning in the next section.